

Proof that $\sqrt{2}$ is Irrational

Assume, for contradiction,

$$\sqrt{2} = 1 + \frac{p}{q} \left| p, q \in \mathbb{Z}^+ \wedge \gcd(p, q) = 1. \right.$$

$$2 = \left(1 + \frac{p}{q}\right)^2$$

$$1 = \frac{2p}{q} + \frac{p^2}{q^2}$$

$$q^2 = 2pq + p^2$$

$$q(q - 2p) = p^2.$$

$$q(q - 2p) = p^2 \implies q - 2p \in \mathbb{Z}^+$$

$$\frac{q - 2p}{p} = \frac{p}{q}.$$

Thus

$$\sqrt{2} = 1 + \frac{p}{q} = 1 + \frac{q - 2p}{p}.$$

Also

$$0 < \frac{p}{q} = \sqrt{2} - 1 < 1 \implies 0 < p < q.$$

Hence $\sqrt{2}$ has a representation with positive denominator $p < q$, contradicting the choice of $\frac{p}{q}$ in lowest terms.

$$\therefore \sqrt{2} \notin \mathbb{Q}.$$

Hence, $\sqrt{2}$ is irrational. ■